Banking Case Project

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Bank Marketing Campaign Analysis

Term Deposit Subscription Study

Background

- This project analyses a real-world banking dataset related to telephonic marketing campaigns conducted by a Portuguese bank.
- The campaigns aimed to promote term deposit subscriptions among customers.

Objective of the Project

- Understand customer demographics and financial attributes
- Analyze campaign-related variables
- Identify patterns influencing term deposit subscription
- Provide data driven insights to improve targeted marketing.

Scope of Analysis

- Data cleaning & preprocessing
- Exploratory Data Analysis (EDA)
- Distribution analysis of key variables
- Correlation analysis with subscription outcome
- Key insights and recommendations.

Dataset Overview

Dataset Summary

- Total Records: **45,211**
- Total Features: **17**
- Target Variable: **y (Term Deposit Subscription)**
- Time Period: May 2008 – Nov 2010

Data Preparation Steps

- Removed 5 duplicate rows
- Dropped redundant columns (marital, day_month)
- Encoded binary variables (yes/no → 1/0)
- Imputed missing values using mode

➢ In this dataset 'Unknown' is a defined category and must be kept, not treated as missing.

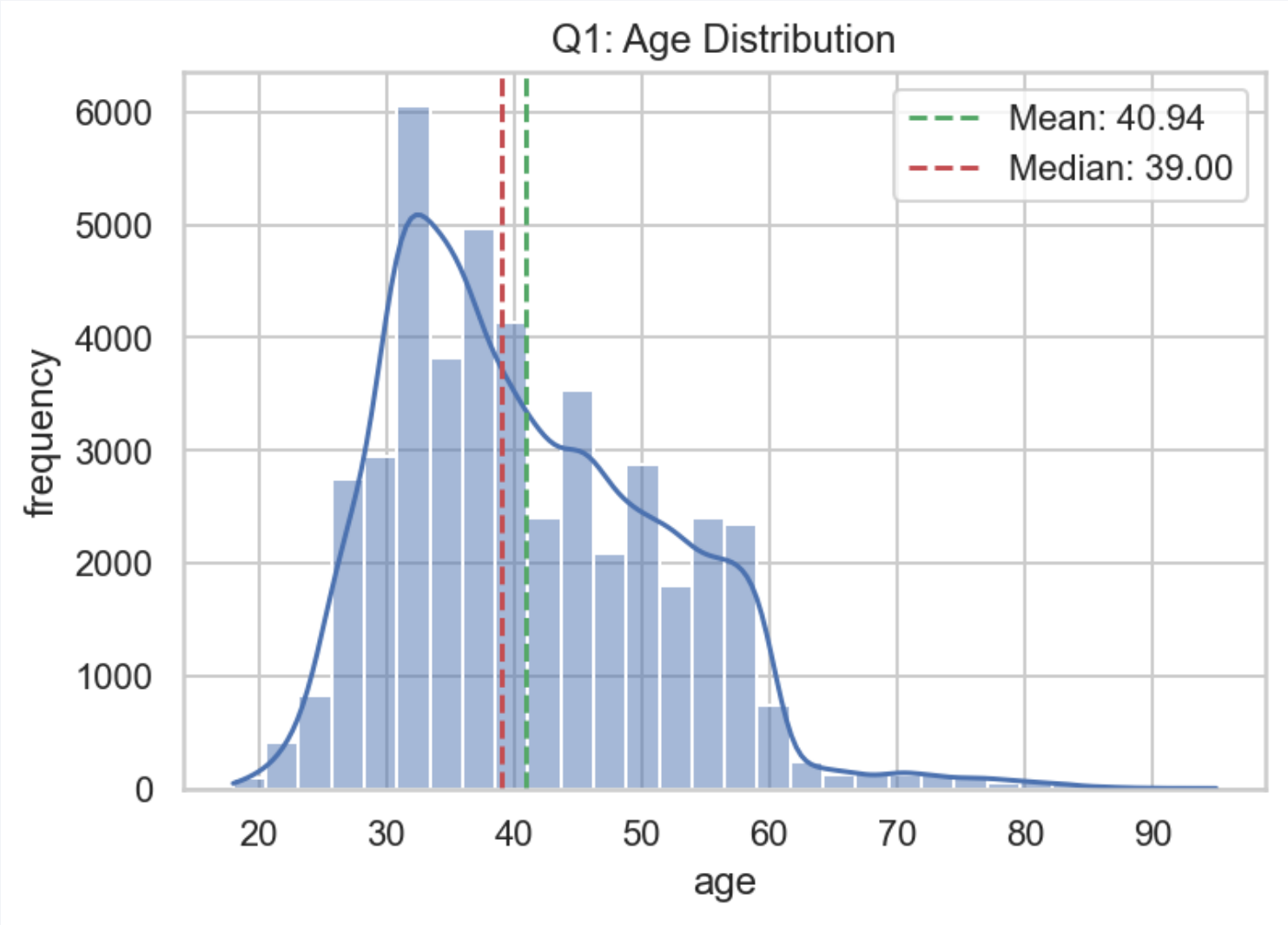
Customer Age Distribution

Key Insights:

- Average age: **40.9 years**
- Majority between **30–50 years**
- Slight right skew
- Few elderly outliers (up to 95 years)

Key Observation:

Middle-aged customers form the primary customer base.



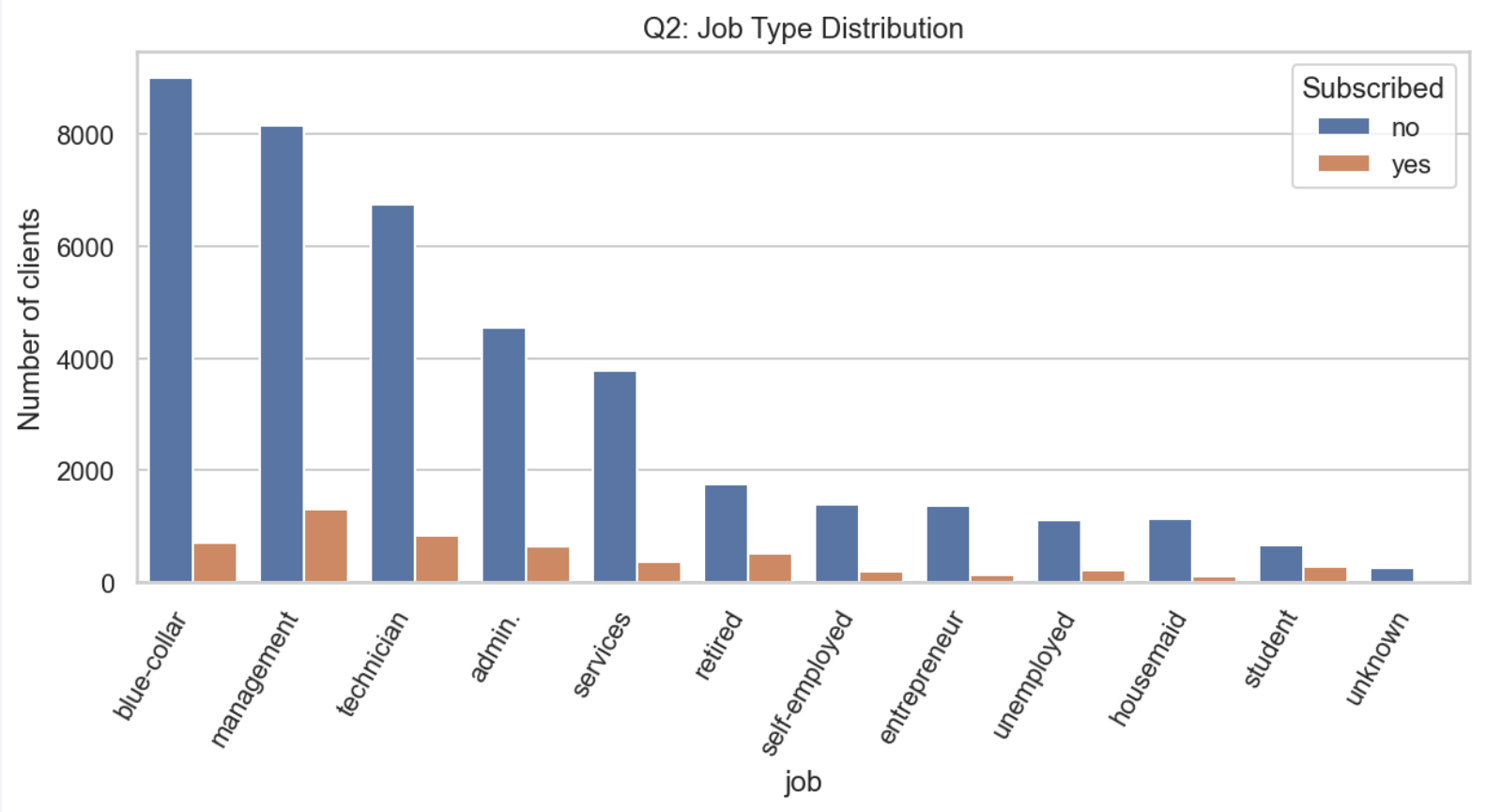
Job Type Distribution

Key Insights:

•Management and technician segments show higher subscription counts.

Most common jobs:
•Blue-collar
•Management
•Technician

•Blue-collar segment has high volume but lower conversion ratio.



Key Observation:

Focus on professional and retired segments for higher campaign efficiency.

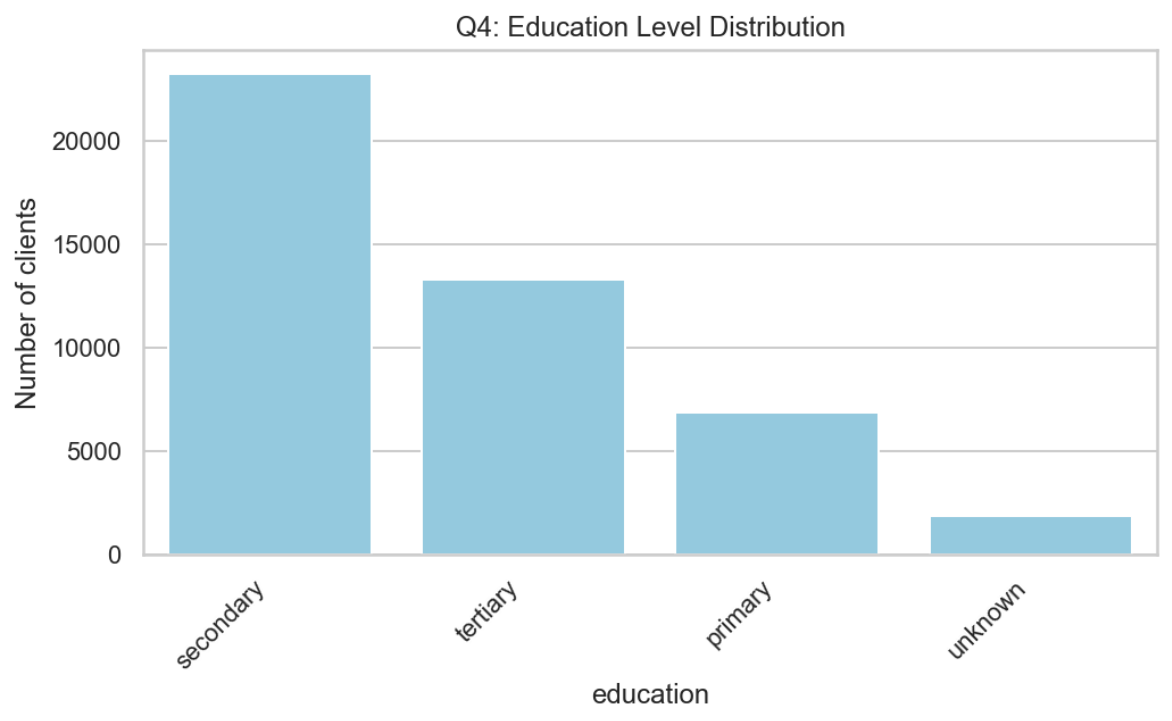
Marital Status & Education Profile

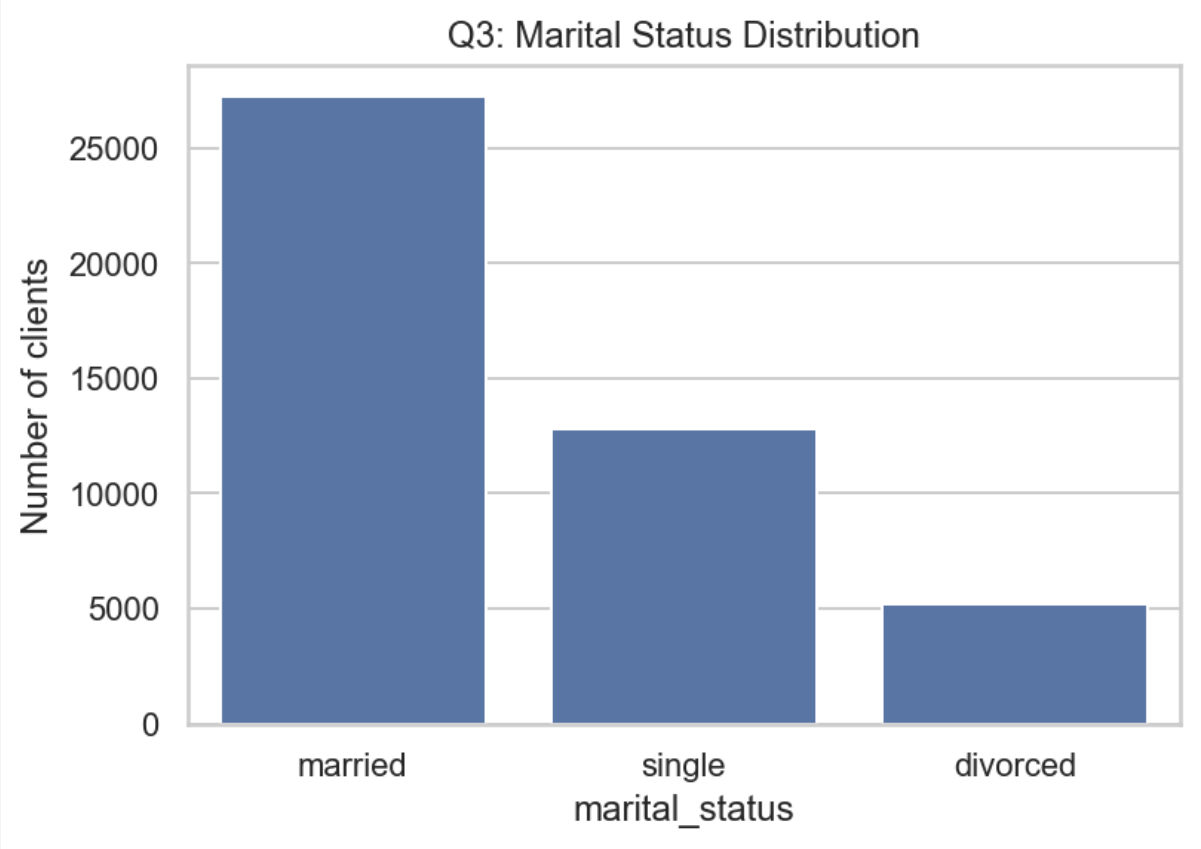
Marital Status:

- 60%+ married
- Singles second largest group

Education:

- Majority have **secondary education**
- Tertiary education is significant

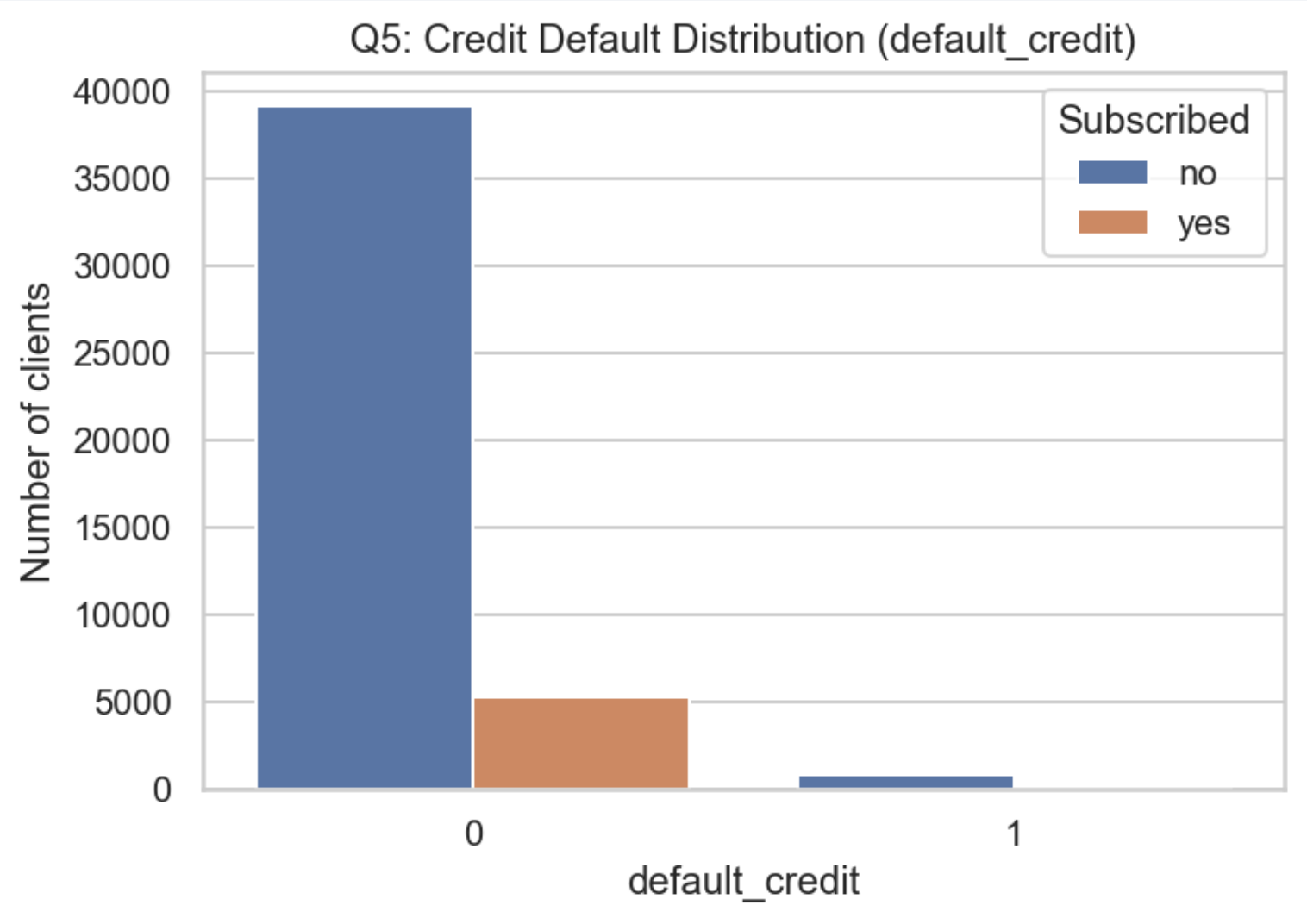




Key Observation:

Target audience mainly married & moderately educated.

Credit Risk Profile



Key Insights:

•Very small percentage (1.8%) in default.

•Clients with credit default rarely subscribe.

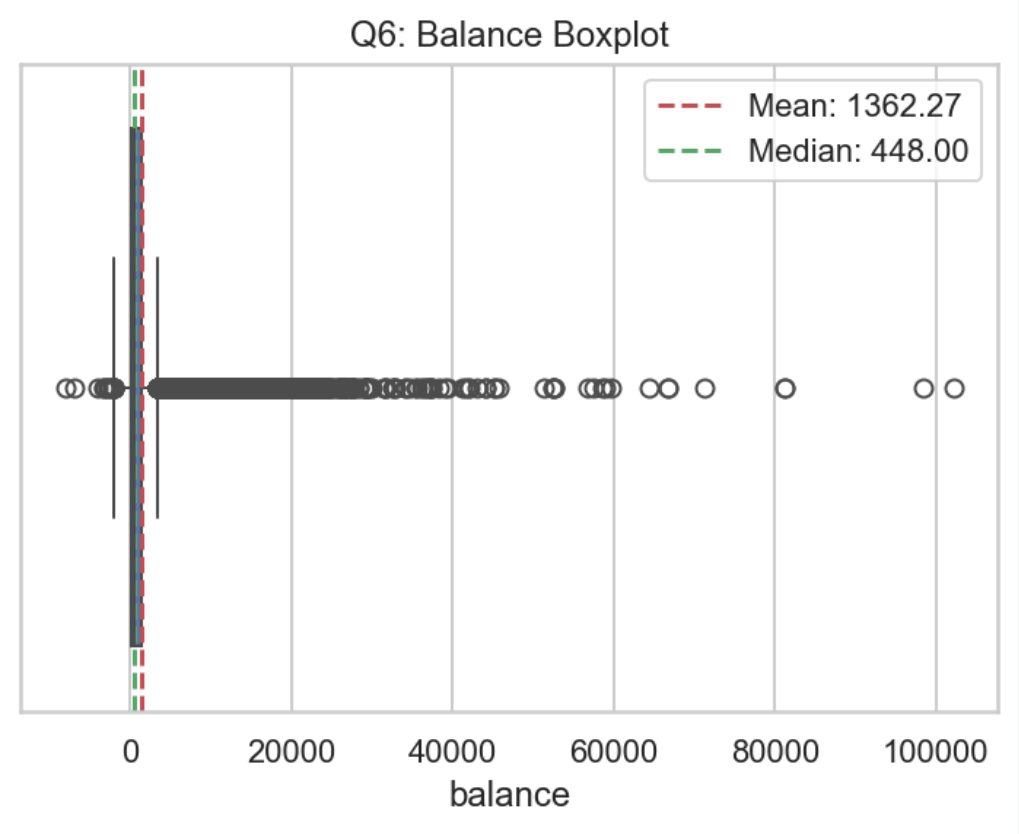
Key Observation:

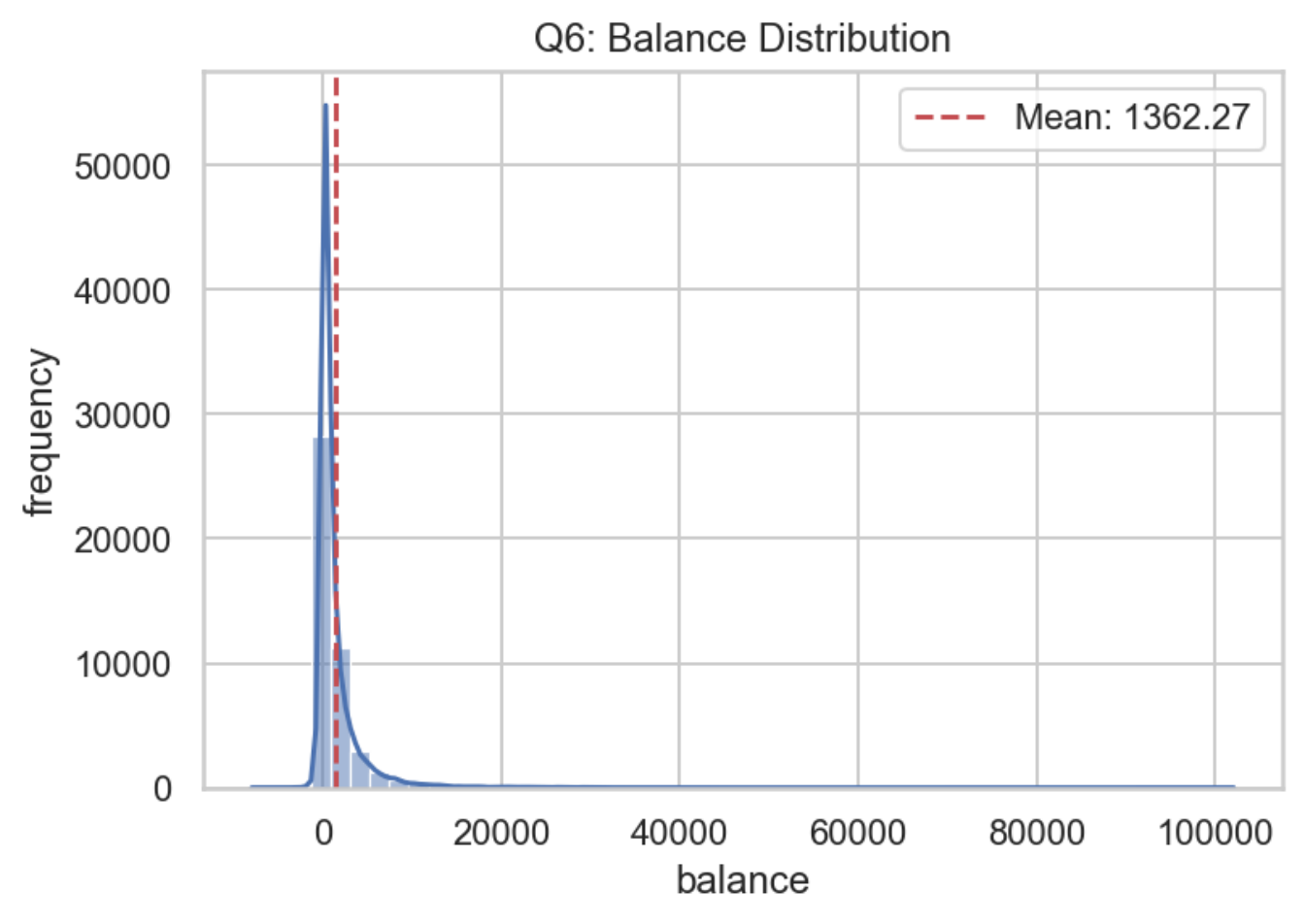
Low-risk customers are primary conversion drivers.

Financial Strength – Balance Distribution

Key Insights:

- Mean balance: **€1362**
- Median: **€448**
- Highly right-skewed
- 4729 balance outliers

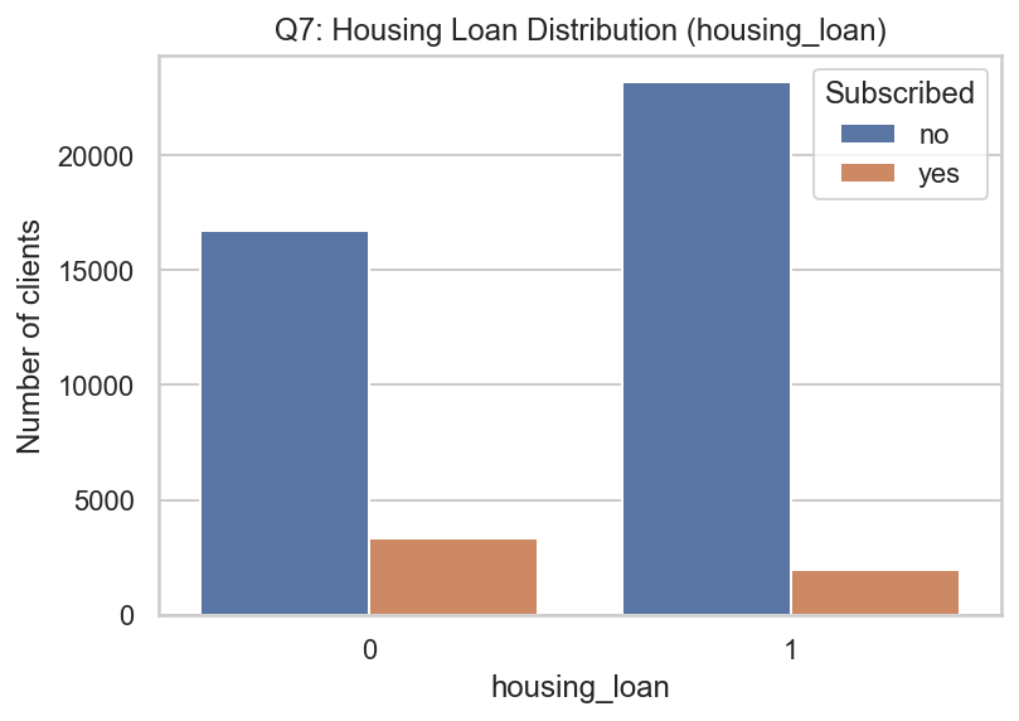




Key Observation:

Small group of high-balance customers skew average.

Loan Status Analysis



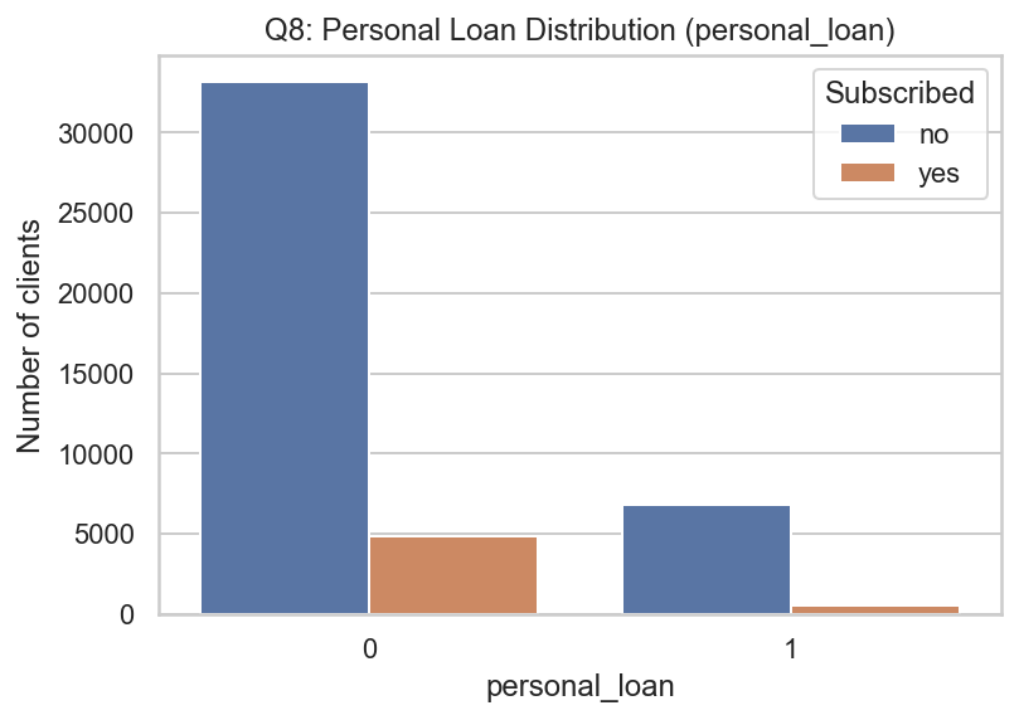
Key Insights:

•Clients without housing loans show slightly higher subscription rates.

•Housing loan holders show lower probability of subscribing.

Key Observation:

Existing financial commitments may reduce investment willingness.



Key Insights:

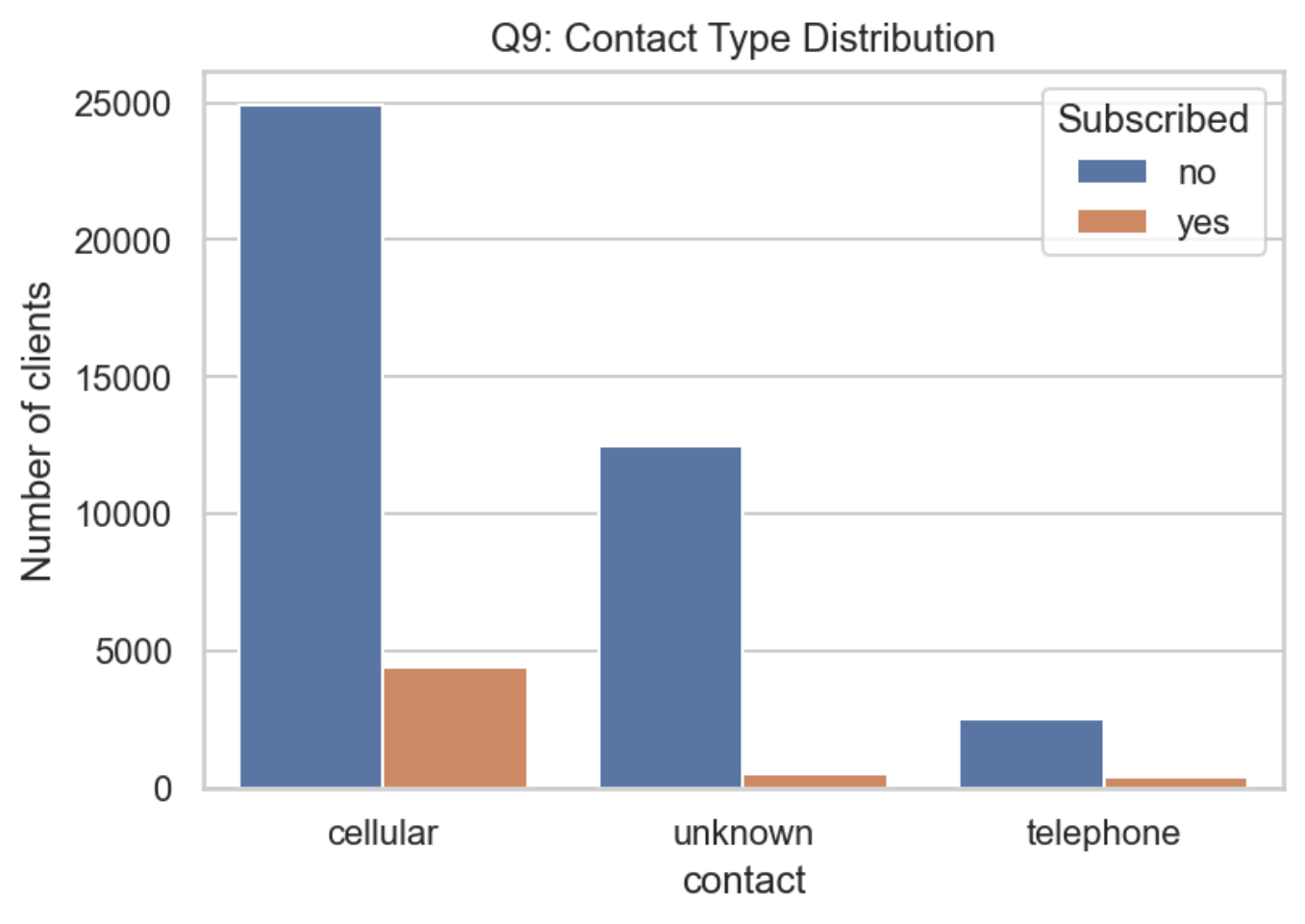
•Personal loan holders show lower subscription rate.

•Non-loan customers convert better.

Key Observation:

Clients already managing debt are less likely to invest.

Contact Method Effectiveness



Key Insights:

•Cellular contact yields highest conversions.
•Telephone channel shows weakest performance.
•Unknown category has poor conversion.

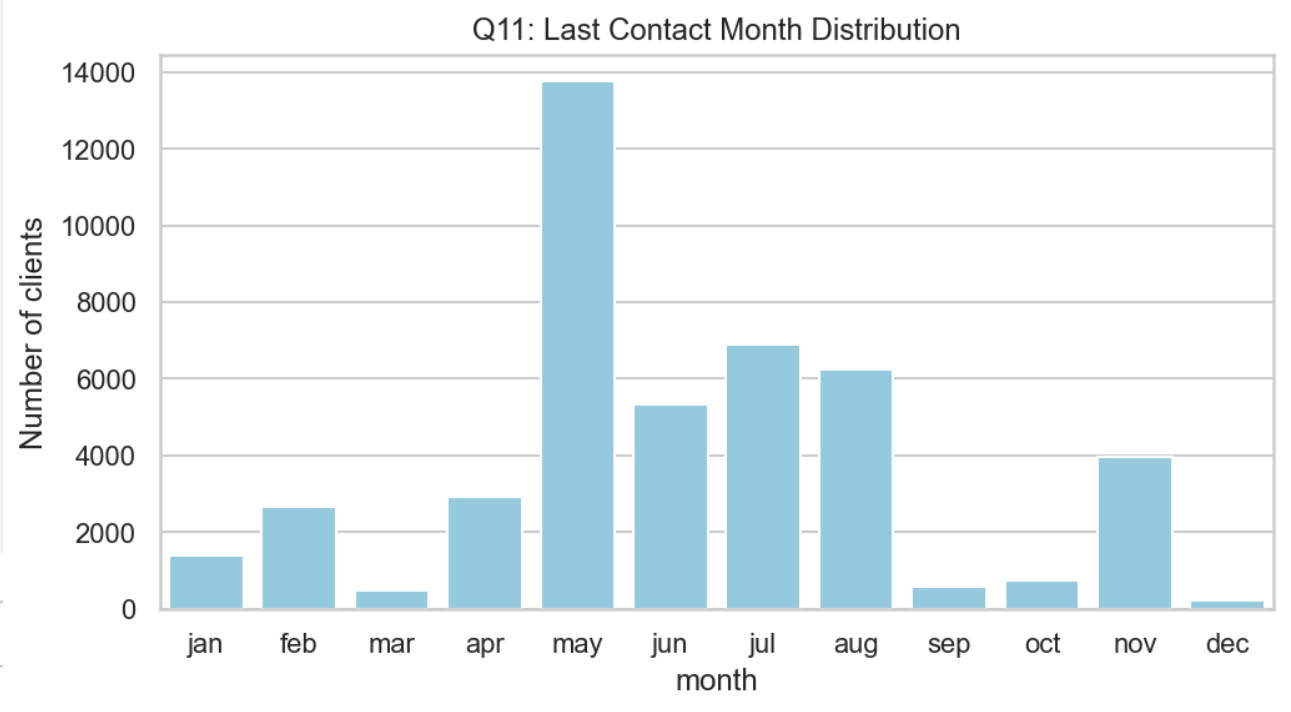
Key Observation:

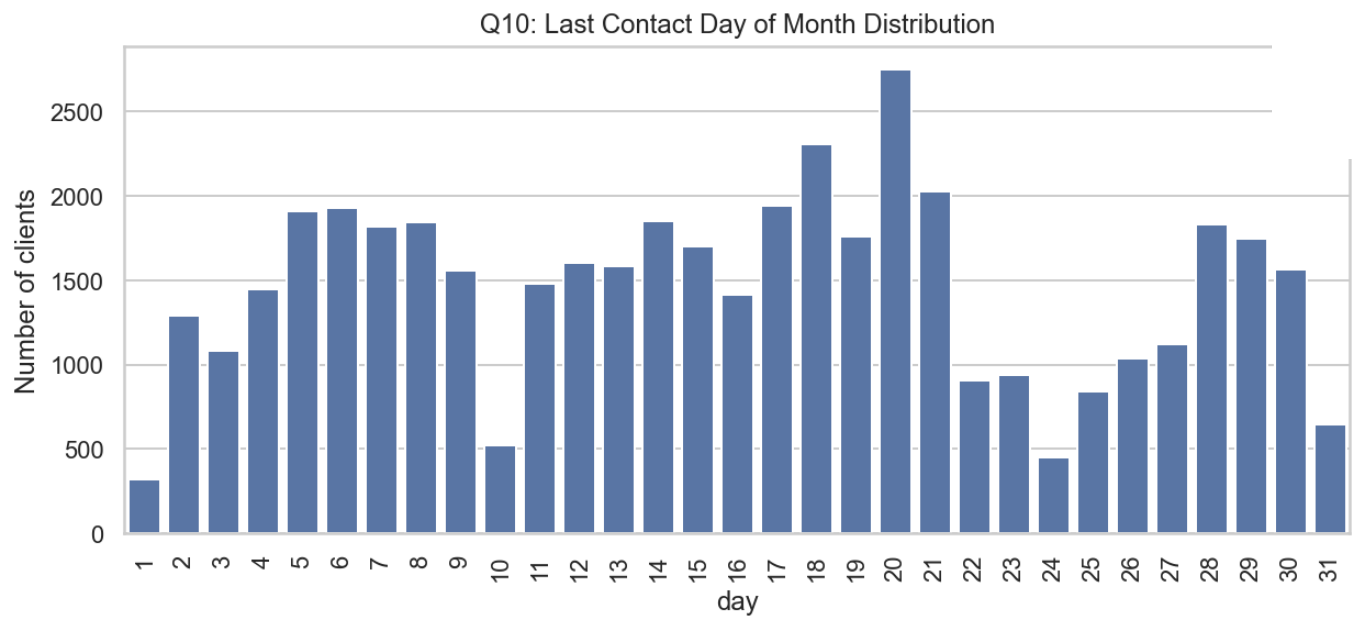
Mobile-based outreach is most effective marketing channel.

Campaign Timing Analysis

Observations:

- Peak campaign month: **May**
- Strong activity in June, July, August
- Day 20 most frequent contact day

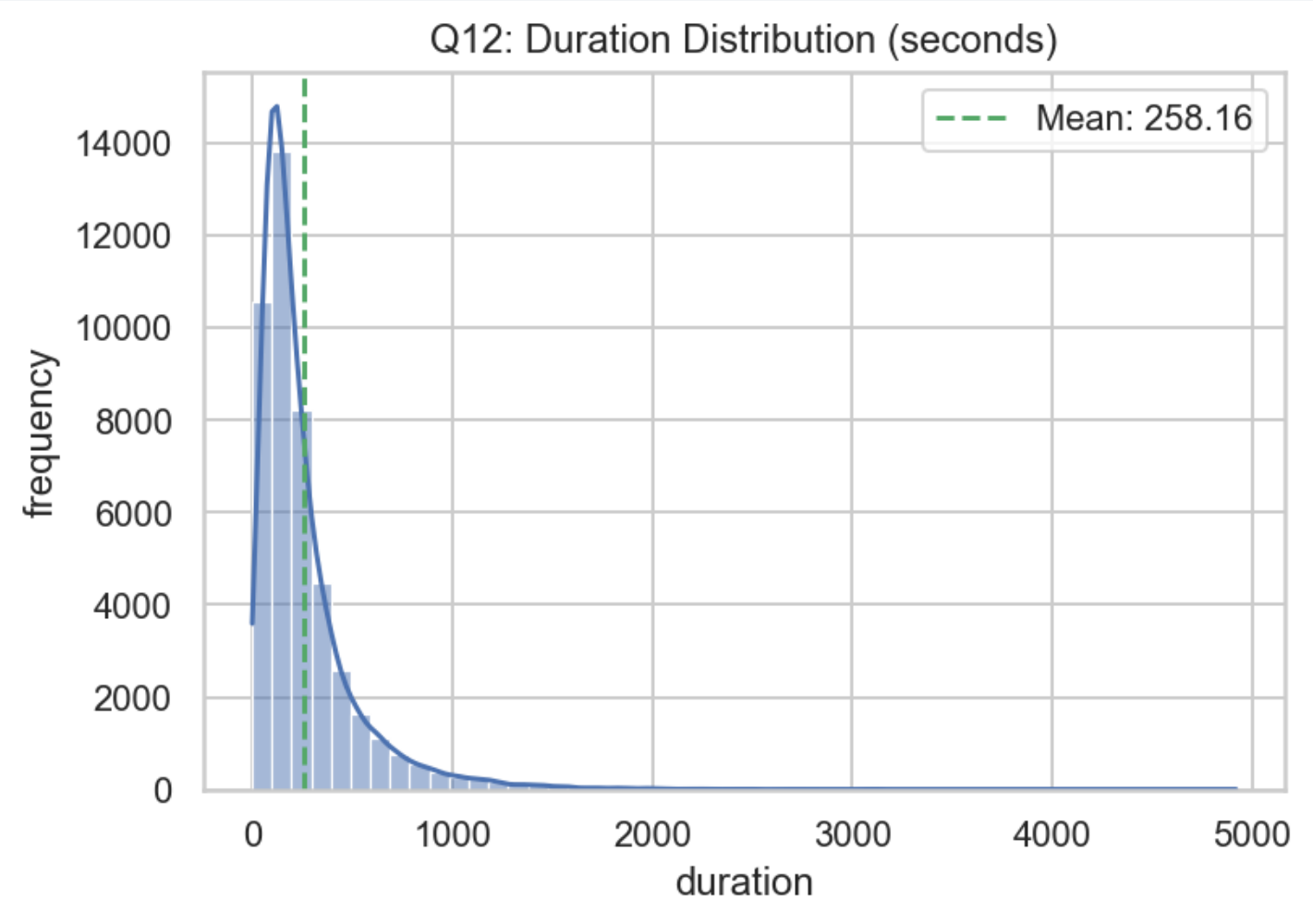




Key Insight:

Campaign concentrated mid-year.

Call Duration Analysis



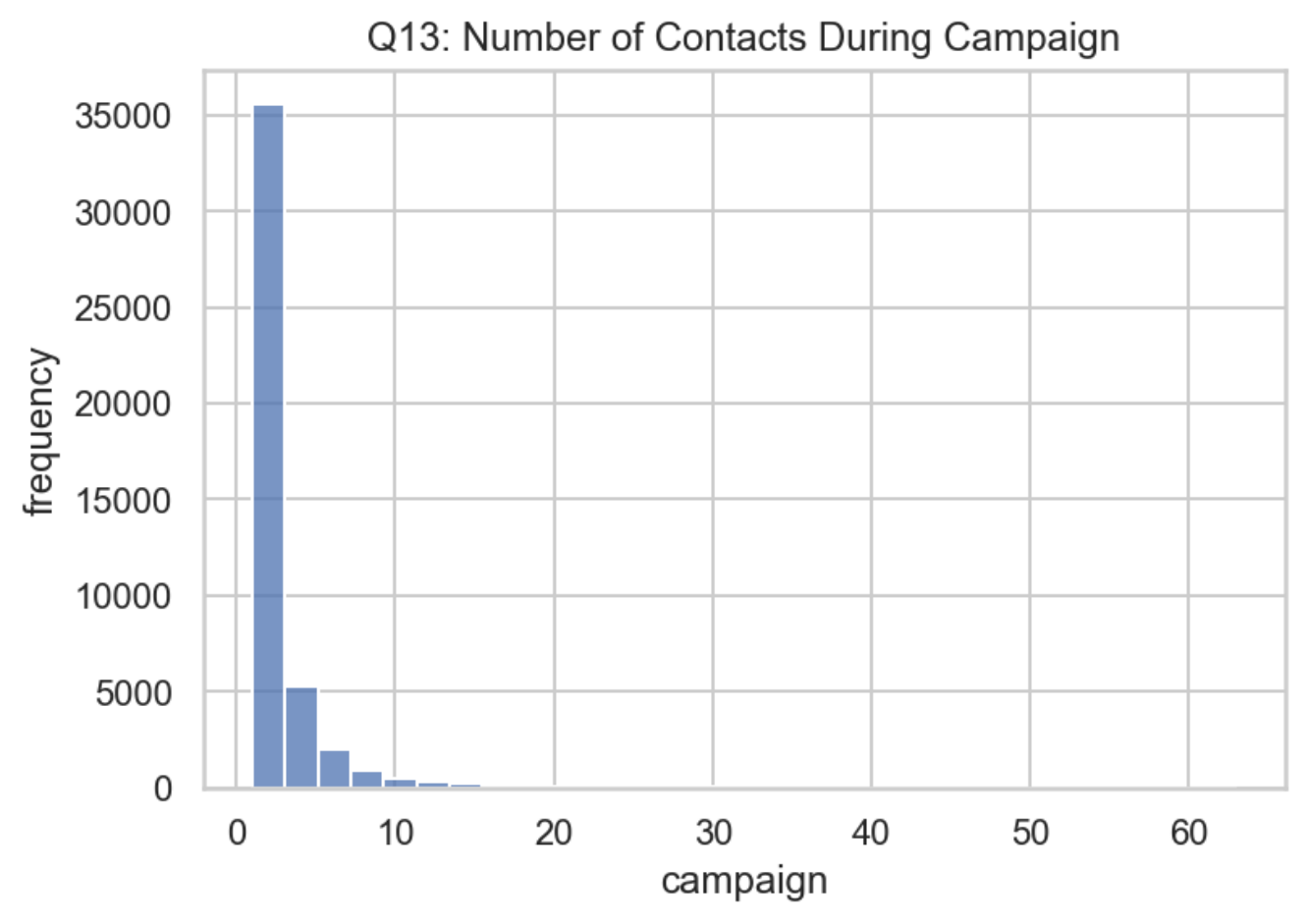
Key Insights:

- Average call duration: **258 seconds**
- Highly skewed distribution
- Long calls associated with higher success

Critical Insight:

Duration has **strongest positive correlation (0.39)** with subscription.

Campaign Intensity



Observations:

- Most clients contacted 1–3 times
- More contacts → lower subscription rate

Correlation:

Campaign correlation with y = **-0.07**

Key Insight:

Excessive calls reduce effectiveness.

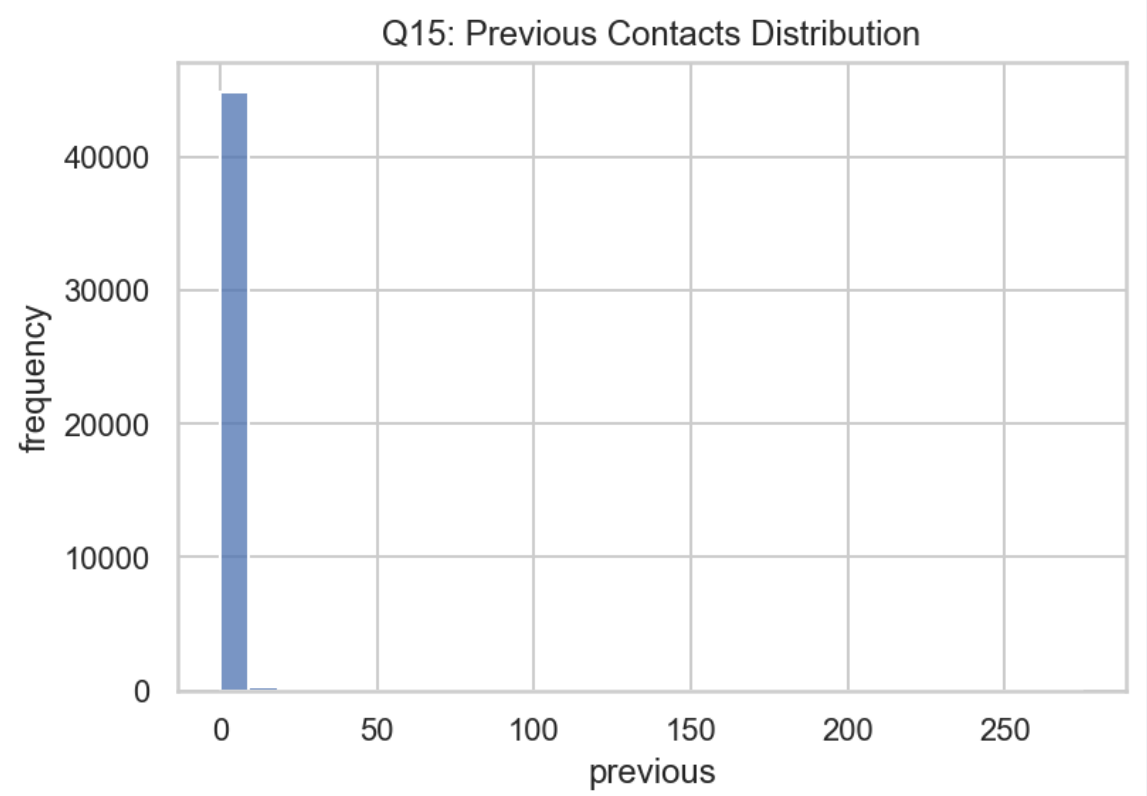
Previous Campaign History

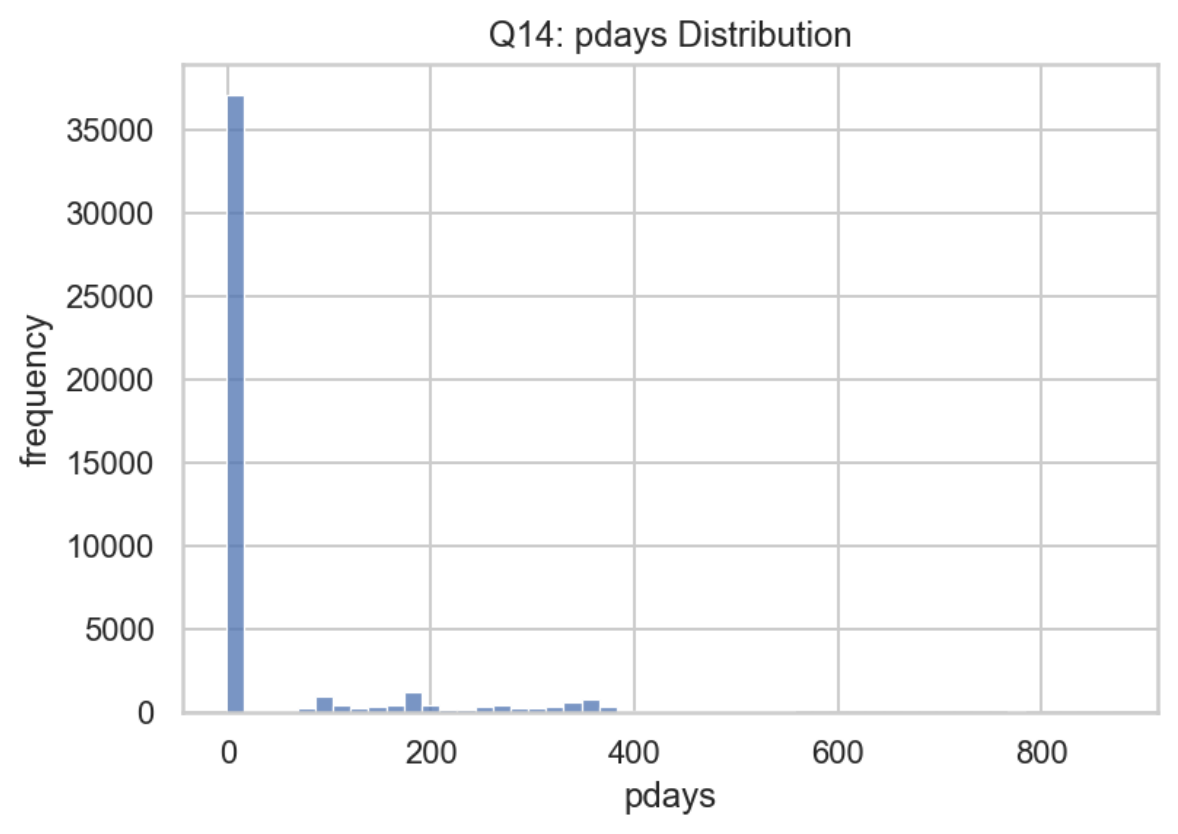
Observations:

- 36,954 clients never contacted before
- Very few had previous interactions

Key Insight:

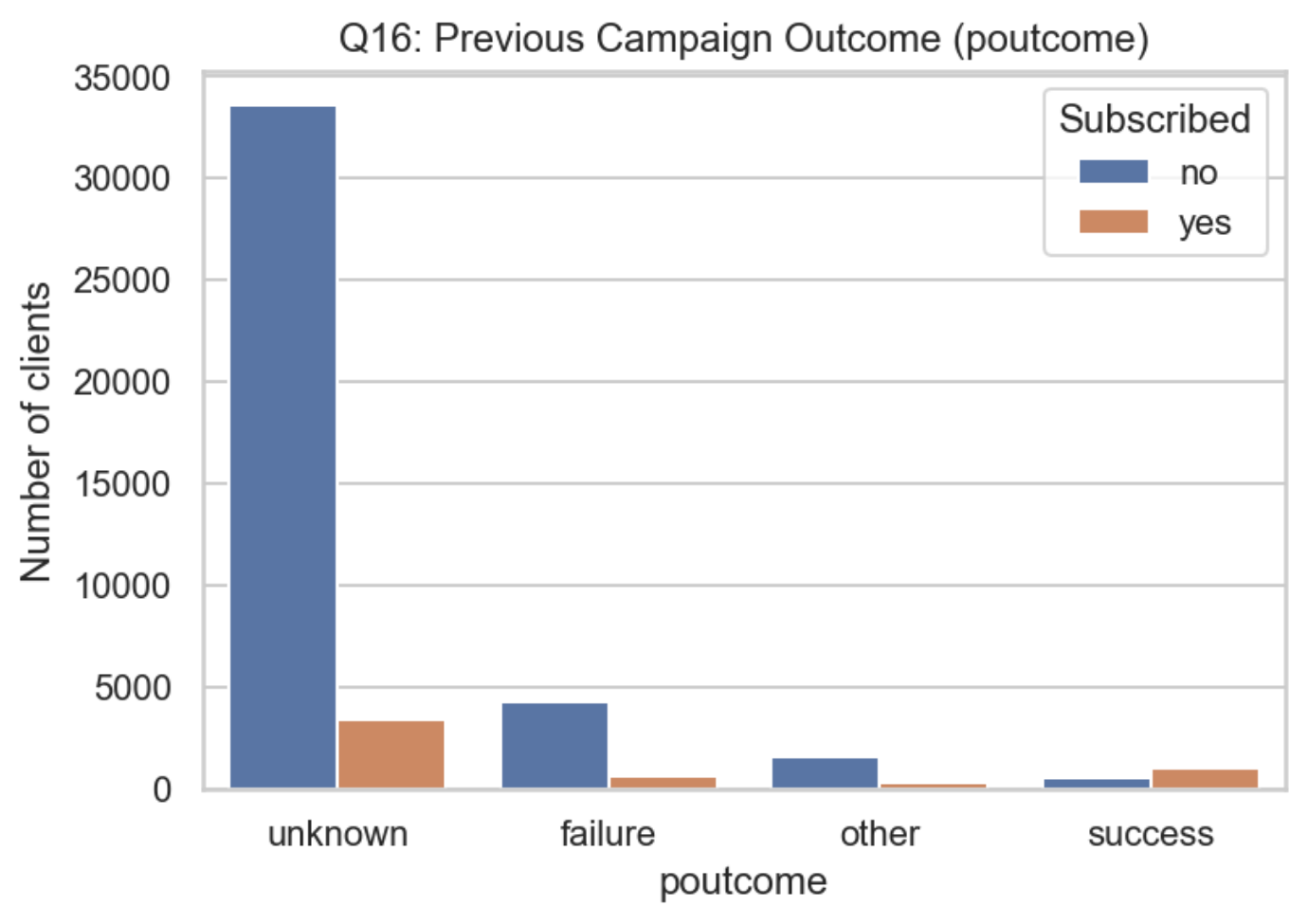
Large untapped customer base.





- The distribution of pdays is highly skewed because many clients were never previously contacted (pdays = -1)
- Similarly, the previous variable is heavily concentrated at 0, indicating most clients had no prior campaign interactions.

Previous Campaign Outcome Impact



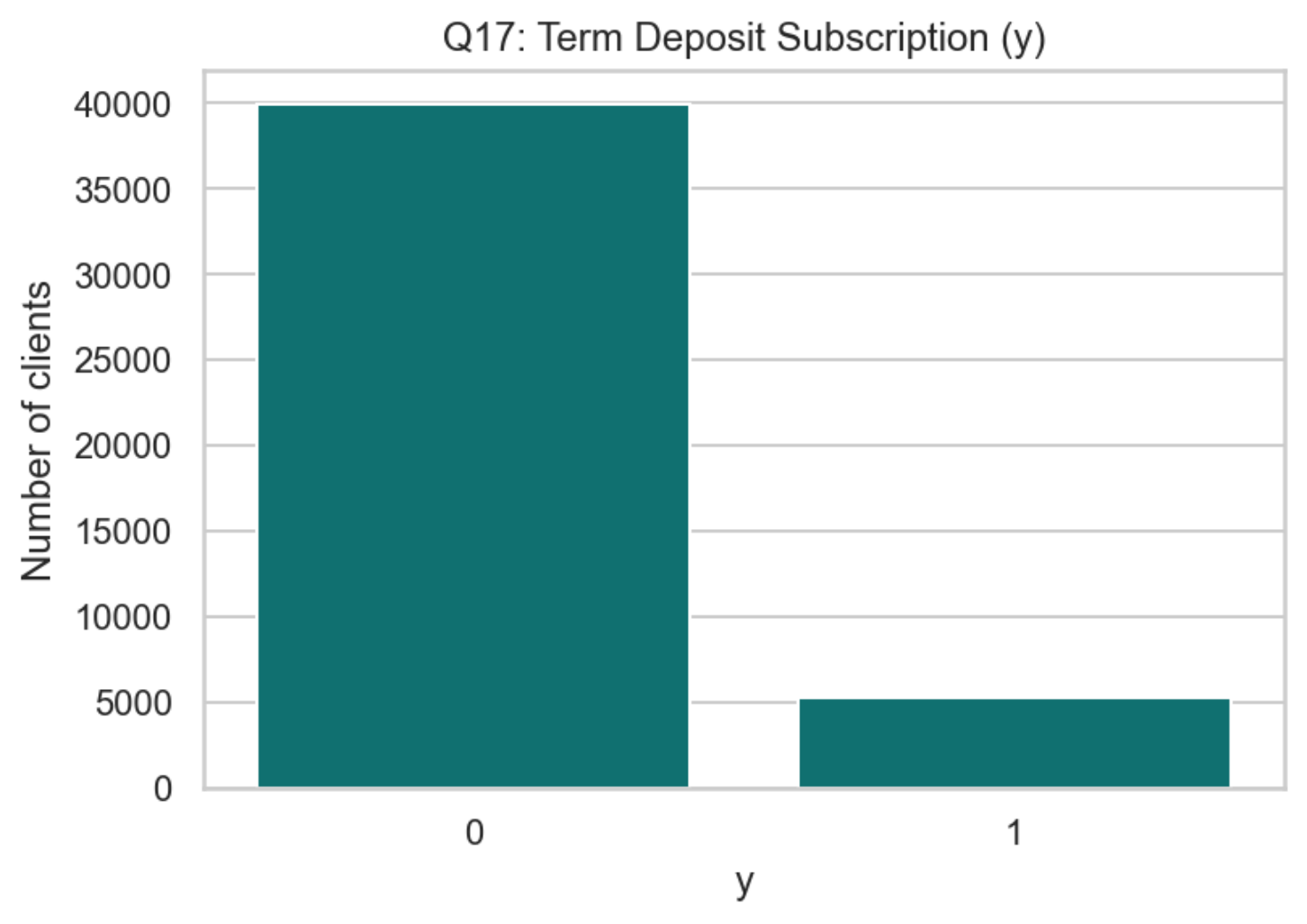
Key Insights:

•Clients with previous success show significantly higher subscription.
•Unknown outcome dominates but has low conversion.
•Previous failure reduces likelihood.

Key Observation:

Past successful engagement is a strong predictor of future success.

Subscription Distribution



Key Numbers:

- Subscribed: **5,289**
- Not Subscribed: **39,922**
- Subscription rate: **11.7%**

Key Observation:

Highly imbalanced outcome.
Campaign conversion is relatively low.

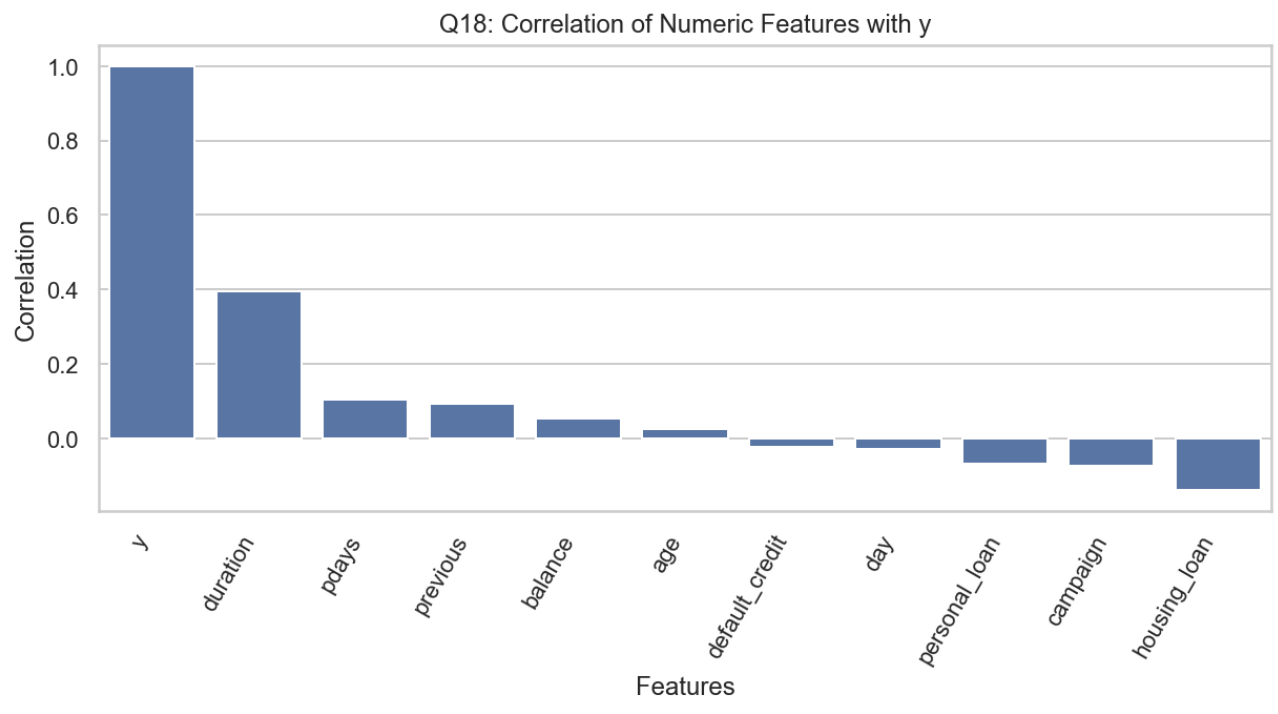
Correlation Analysis

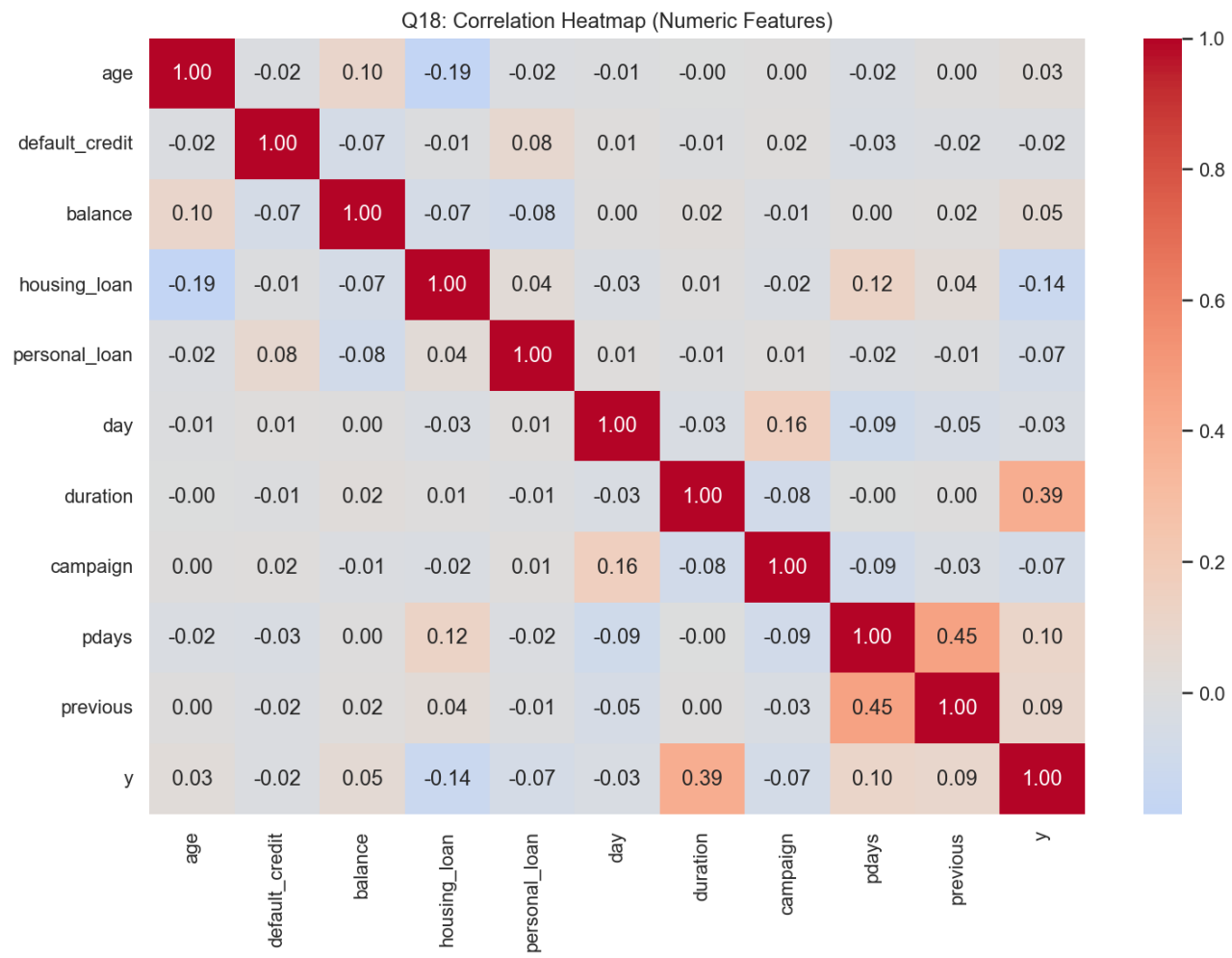
Strongest Positive Drivers:

- Duration (0.39)
- pdays (0.10)
- previous (0.09)

Negative Drivers:

- Housing loan (-0.14)
- Campaign (-0.07)
- Personal loan (-0.07)





Key Insight:

Call duration is the strongest driver of subscription probability.

Strategic Recommendation

- **Focus on Call Quality, Not Quantity**
  Longer, meaningful conversations increase conversion.
- **Prioritize Customers with Past Success**
  Higher likelihood of subscription.
- **Optimize Campaign Frequency**
  Avoid excessive repeated calls.
- **Target Mid-aged, Working Professionals**
  Core customer segment identified.

Thank You